

Enhancing Glass-to-Glass Bonding at Room Temperatures Using Laser Surface Modification With Ti/Cu/Ti/TiO₂ Thin Films

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Abstract—There are many applications for glass-to-glass bonding, such as manufacturing optical devices, electronic components, and microsystem applications. This article presents a simple and effective method for glass-to-glass bonding using laser modification with Ti/Cu/Ti/TiO₂ thin films. Ti/Cu/Ti thin films were prepared on glass using magnetron sputtering and then exposed to the atmosphere for 30 min at room temperature for preoxidation to obtain Ti/Cu/Ti/TiO₂ thin films. The TiO₂ thin film was intentionally formed to improve the bonding quality. To analyze the bonding mechanism, the energy dispersive spectroscopy (EDS) and X-ray photoelectron spectra (XPS) analyses studied the microstructural evolution. The results show that Si diffuses from the glass to the thin films by laser irradiation. The laser beam penetrated the glass and was directly absorbed by the TiO₂ thin films. Ti diffuses from the oxide thin film toward the glass, forming Ti–Si covalently structured silicate (the main component is TiSi₂) in the bonded region. The formation of TiSi₂ is the critical process for this laser bonding. Furthermore, the bonding strengths corresponding to different contact pressures applied during the bonding process were compared and analyzed. The results show that the method achieves more robust glass/TiSi₂/Ti/Cu/TiSi₂/glass bond pairs with tensile strengths up to 14 MPa and low bonding defect density (<0.05%). Note that this laser bonding technology was processed at room temperature, allowing the use of thermally sensitive materials in device manufacturing. To verify the application potential of the laser bonding technique in 3-D glass packaging, a simple coplanar waveguide (CPW)–through-glass vias (TGVs)–CPW interconnect structure was developed on three-layer glass interposers in this article. The interconnection loss was successfully measured by constructing a three-layer interconnect structure with laser bonding. The

results show that the insertion loss can meet the requirements of glass-based microsystem applications.

Index Terms—Bonding strength, contact pressures, glass-based microsystem, laser modification, room temperature.

I. INTRODUCTION

GLASS interposers have great potential for high-frequency and high-speed communications applications due to lower microwave loss and processing costs [1], [2], [3], [4], [5], [6], [7], [8]. Micromachining the glass interposers to create microchannels and fabricate through-glass vias (TGVs) constitutes a pivotal milestone [2], [3], [4], [5]. Besides the laser ablation, CF₄-based plasma etching, new cost-effective electrochemical discharge machining (ECDM)-based methods to etch glass have emerged [2], [4]. Notably, TGV-based 3-D microstructures can be fabricated in glass by a lithography-free process [5]. TGV technology applied to glass interposers is crucial for realizing multilayer stack interconnects. For example, the reported glass-based multilayer package antennas [7] and inductors [4], [5], [8].

Bonding is an indispensable technology in the 3-D packaging process [7], [9], [10], [11], [12], [13], [14]. Since it allows bonding multiple layers of glass interposers, devices to interposers, together while providing the necessary electrical connections and mechanical support [10], [11], [12], [13], [14], [15], [16], [17]. Currently, popular bonding techniques in glass packaging include direct bonding of glass to metal [18], [19], [20], laser bonding between glass and silicon [9], [21], glass-to-glass bonding using glass frit [22], [23], and metal thermocompressive bonding [24], [25], [26], among others. Each of these methods presents distinct advantages and drawbacks. Direct bonding of metal and glass can achieve high bonding strength and reliability but has problems such as extensive bond spots that need to be more easily integrated and high laser energy that causes glass breakage [19]. Laser bonding between glass and silicon can produce good bonding strength and gas sealing. However, it is unsuitable for bonding with surface-structured metals, and there still needs to be more understanding of the electrical properties postbonding [21]. Glass-to-glass bonding using glass frit is ideal for extensive area bonding and offers

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excellent hermeticity. However, the bonding quality is heavily influenced by the volume and homogeneity of the glass frit particles, necessitating accurate alignment and pressure control [22]. Metal thermocompressive bonding has been widely used in manufacturing advanced electronic devices due to its good gas sealing and electrical properties [27]. Unfortunately, metal thermocompressive bonding conditions require a heated environment [18], which is not conducive to integrating temperature-sensitive components. Moreover, symmetrical bonding surface geometries are required, which is not conducive to cost control, and high-precision alignment requirements.

This article introduces the laser modification of Ti/Cu/Ti/TiO₂ thin films using laser bonding technology to achieve bonding between multilayer glass. While ensuring good bonding strength and gas sealing, it was applied to multilayer surface geometries bonding, which possesses good electrical properties. This article investigates the changes in valence and compositional characteristics of the elements on the thin films and glass surfaces before and after bonding using characterization techniques such as XPS, EDS, and Raman spectra and analyzes the mechanism of laser bonding. Furthermore, the bonding strength under different contact pressures was tested to evaluate the bonding quality. Considering the glass packaging multilayer interconnect, the method was compared with traditional thermocompressive bonding to verify its low-temperature and low-cost advantages in the glass packaging process. Finally, a simple coplanar waveguide (CPW)–TGVs–CPW interconnect structure was developed on three-layer glass interposers, and the interconnect loss of TGVs was measured. The results show that the requirements of glass-based microsystem applications can be met.

II. DEVICE FABRICATION

The films were prepared by magnetron sputtering equipment, and a layer of Ti thin film with a thickness of 200 nm was sputtered on the glass surface. The sputtering atmosphere was 60-sccm argon, sputtering pressure was 1 Pa, sputtering power was 150 W, and sputtering time was 5 min. The copper thin film was prepared under the same sputtering conditions with a sputtering time of 30 min, and 800-nm copper thin film was obtained. Then, the Ti thin film was sputtered for 5 min under the same conditions to get a 200-nm Ti thin film again. Finally, Ti/Cu/Ti thin films were preoxidized by exposure to the atmosphere at room temperature for 30 min to obtain Ti/Cu/Ti/TiO₂ thin films.

Fig. 1(a) and (b) shows the schematic of the laser bonding. The glass substrate of the Ti/Cu/Ti/TiO₂ thin films was placed on the bottom of the bonding device, and another piece of polished glass was placed on top of the thin films. The bonding device was customized for the specific dimensions of the glass substrate (50.8 × 50.8 × 0.65 mm). The glass substrate was positioned correctly and then fixed with screws and pressurized. A film pressure sensor was placed at the bottom of the bonding device to real-time monitor the contact pressure. Finally, the device was placed on the laser bonding machine. The laser energy was 3 W, the focal length was 0.3 mm,

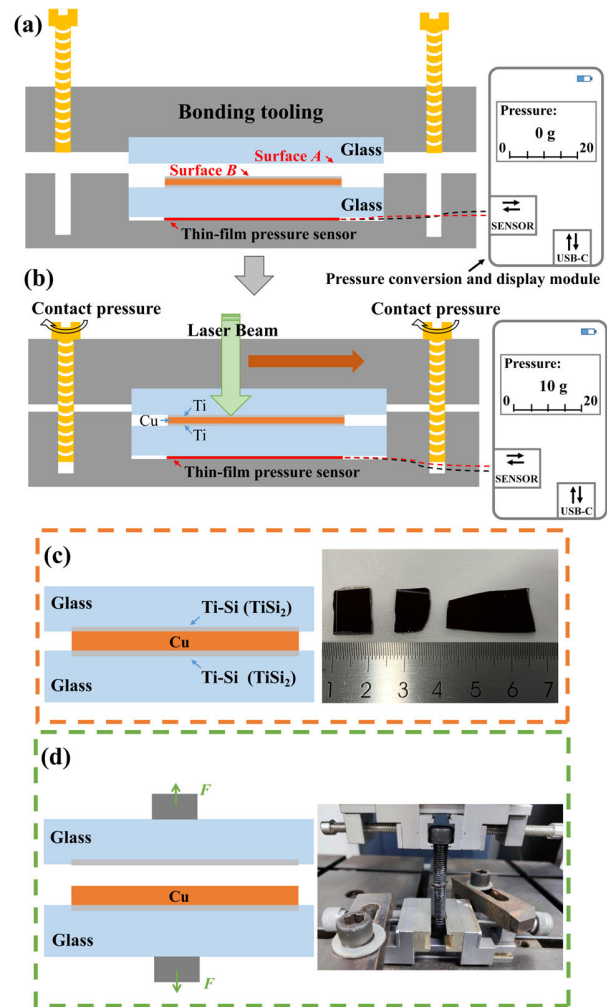


Fig. 1. Laser bonding preparation process and tensile strength testing. (a) Laser bonding preparation process, (b) laser bonding process, (c) samples after laser bonding, and (d) bonding strength test process.

and the machining speed and acceleration were 5 mm/s and 50 mm/s³, respectively. The wavelength of the laser is 355 nm, and the spot size is 10 μm.

The microstructures and EDS line scan of the bonded area cross-sectional were observed by scanning electron microscopy/energy dispersive spectroscopy (EDS). X-ray photoelectron spectra (XPS) were tested by Al-Kα (1486.6 eV) radiation. Binding energies calibrated concerning C (1s) at 285 eV were accurate within ±0.2 eV. Fig. 1(c) shows the schematic of the cross section and the sample after laser bonding. Since the bonding strength test requires cutting the sample into small pieces, this article was limited by the experimental conditions and only used a gold steel knife to cut the sample. As shown in Fig. 1(d), the bonding strength of the samples was tested using a microcomputer-controlled electronic universal testing machine.

The electromagnetic simulation High-Frequency Structure Simulator (HFSS) constructed a simple CPW–TGVs–CPW interconnect structure model. The S-parameters were measured up to 40 GHz utilizing 150-μm pitch ground-signal-ground (GSG) probes (High-Performance Microwave Probes) connected to a vector network analyzer. The vector network

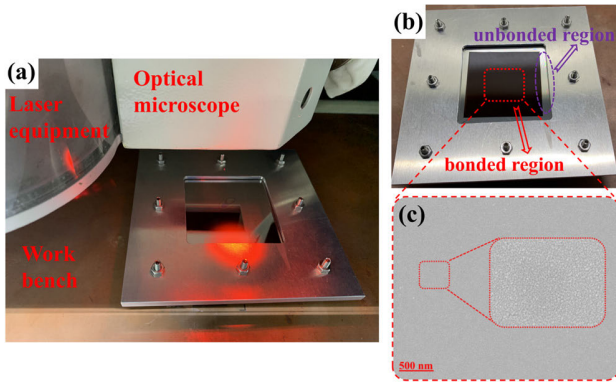


Fig. 2. Laser bonding equipment. (a) Experimental setup of the laser bonding process. (b) Samples after laser bonding with the bonding device. (c) SEM image of the bonded area surface.

analyzer used for this study was an Agilent PNA N5247A, capable of measuring frequencies ranging from 10 MHz to 67 GHz. For more details on the fabrication process of the three-layer interconnect structure, please refer to [28].

III. RESULTS AND DISCUSSION

Fig. 2(a) shows the actual laser bonding device. Note that the upper glass layer is transparent, and in practice, it is necessary to focus the laser directly on the TiO_2 thin film by adjusting the laser focal length. The laser provides sufficient energy and opportunity for the diffusion of Ti. Thanks to the light transmission of the upper glass, which is shown in Fig. 2(b) and (c) shows the morphology of the bonded region. Fig. 2(b) shows the bonded sample with CuO color in the bonded region and argentite in the unbonded region (considered the Ti and TiO_2 thin films). This indicates that during the laser bonding process, the TiO_2/Ti layer mainly absorbs the laser energy and combines with the glass to appear transparent. The unbonded region, however, remains prebonding. Furthermore, as shown in the SEM image in Fig. 2(c), no bond spots were observed in the bonded region, which is different from the case reported in previous studies [18], [19], [20], [21]. This may be attributed to the direct fusion process of the Ti layer with the glass after dynamically absorbing part of the laser energy. If the bond spots are too large, they are prone to problems such as unstable metal film resistance changes and poor contact, which can lead to the malfunctioning of the microsystem package [29], [30].

The cross section of the bonded region [Fig. 3(a)] and the unbonded region [Fig. 3(b)] shows that the upper Ti and TiO_2 layers have disappeared, forming a bonded region fused with glass. The laser energy also affected the lower Ti layer to form a bonded region with the lower glass layer. In addition, the thickness of the Cu film in the bonded region increased, which may be related to the diffusion behavior of Cu by the laser energy. The EDS line scan image at the glass-films-glass cross section is shown in Fig. 3(c). Based on the distribution of Cu, it can be further determined that Cu absorbs part of the laser energy and produces diffusion behavior. The images of the elemental distribution of Ti and Si show that, as expected, Ti absorbs the laser energy and diffuses it toward the glass. Meanwhile, Si was also affected

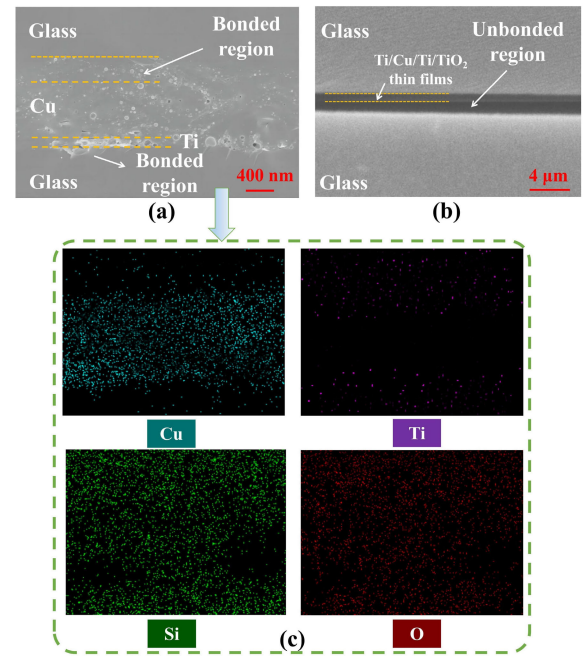


Fig. 3. Cross-sectional SEM image of (a) bonded area and (b) unbonded area, and (c) SEM image with corresponding EDS mapping images of elemental of Cu, Ti, Si, and O.

by the laser energy toward metal thin films. Ti and Si form a new covalent structure at the bonded surface. The laser energy also included a small part of the bonded region in the bottom glass. Due to the immediate and fast reaction of the laser beam, only very little heat was transferred to the surrounding materials. Therefore, the temperature of the entire bonded region increases only slightly [12]. Moreover, the heating and cooling times during the laser bonding process are short. This method avoids localized overheating (generation of thermal stresses) and crystallization of the metal films. It is suitable for materials requiring low-temperature bonding, such as plastics and heat-sensitive materials. Especially for glass-based packaging microsystems, the remaining copper layer in the bonded region can be utilized directly as a redistribution layer (RDL), such as the grounding layer of antenna-in-packaging [7]. The patterning process can be implemented now by adjusting the laser paths, greatly simplifying the process steps.

Fig. 4 shows the distribution of the elements in the bonded region. EDS scans were performed at five different positions in the bonding region area. The results are as expected, mainly Ti and Si elements in the upper bonded region. Since the lower Ti thin film was also affected by the laser energy, it formed part of the bonded region with the glass. Note that the bonding regions in Figs. 3(a) and 4 show the presence of voids. These voids' forms can be attributed to the mismatch of coefficients of thermal expansion (CTEs) between the glass and the Ti and Cu thin films. Thermal stresses occur during laser bonding, and stress concentrations occur in the bonding regions between the glass and the metal films. These stress concentration regions cannot withstand the higher stresses, forming voids on the bonding surface. The formation of voids can hurt the bonding strength. First, voids reduce the effective contact area in the bonding regions, reducing the bonding strength. Second, voids

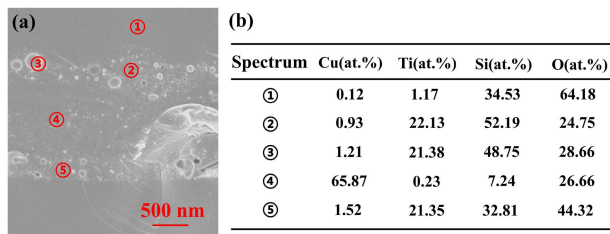


Fig. 4. Structural characteristics of the bonded region. (a) Cross-sectional view of the bonded region with the position of EDS spot and (b) corresponding composition.

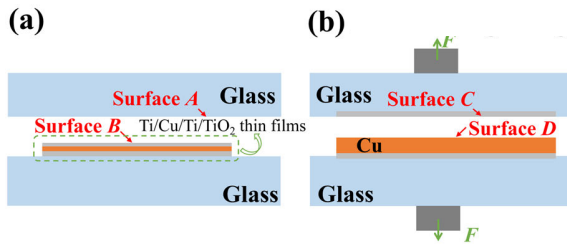


Fig. 5. Schematic of different characterization surfaces. (a) Glass surface (Surface A) and thin film surface (Surface B) before laser bonding. (b) Surfaces C and D after laser bonding strength test.

also form stress concentration areas in the bonding area, which reduces the strength and durability of the laser bonding. Also, the voids will act as the initiation point of stress concentration, which may lead to the formation and expansion of cracks and eventually lead to bonding failure. Therefore, minimizing the creation of voids during the laser bonding process is crucial. Kannoja and Dixit [25] introduced that the generation of voids during the bonding process can be effectively reduced by preannealing, increasing the contact pressure and reducing the bonding surfaces' roughness.

Preoxidation of the thin films' surface before laser bonding to obtain TiO_2 thin film can improve the metal surface's chemical activity and increase the chemical affinity between the metal thin film and the glass, thus improving the bonding strength and quality of laser bonding [18]. To facilitate the reader better understand the comparison of the characterization of different surfaces later on, Fig. 5 shows different surfaces named before laser bonding and after the bonding strength test. The glass and thin films surface before bonding are named Surfaces A and B, respectively, and after bonding, they are correspondingly named Surfaces D and C, respectively.

The surface roughness of the bonding areas plays a vital role in the bonding quality. To facilitate comparison, two sets of samples with different surface roughness were used in this article, and the two other roughness surfaces were measured using AFM, as shown in Fig. 6. For Sample A, the root-mean-square (rms) R_q of roughness and the arithmetic mean deviation R_a of the profile for Surface A were 1.094 and 0.327 nm, respectively. R_q and R_a of Surface B were 47.6 and 29.2 nm, respectively. For Sample B, R_q and R_a of the profile for Surface A were 1.533 and 0.632 nm, respectively. R_q and R_a of Surface B were 59.7 and 39.9 nm, respectively. These two samples' bonding strength (tensile strength) test results are shown in Fig. 7(a). Sample A's tensile strength is more significant than that of Sample B's under the same laser bonding environment. This is attributed to the

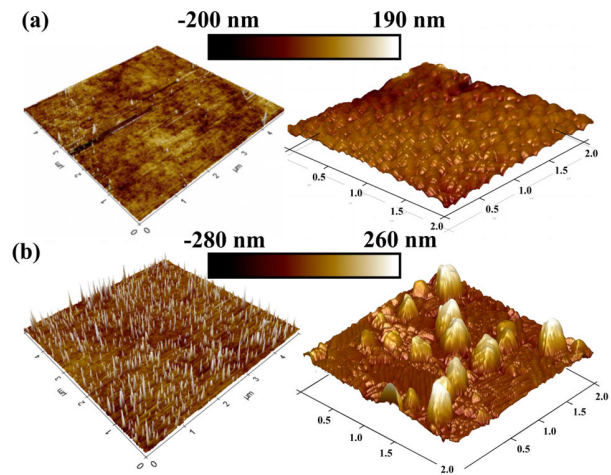


Fig. 6. AFM image Surfaces A and B of (a) Sample A and (b) Sample B.

uneven metal surface, which reduces contact with the glass, resulting in incomplete bonding (voids) [19], [20]. Simultaneously, the diffusion process during laser bonding is influenced by uneven temperature distribution, subsequently impacting bonding quality and device performance [19]. Furthermore, nonuniform pressure from the rough surface traps gases during bonding, weakening bonds in specific areas and, thus, overall bonding quality [19], [20].

To investigate the laser bonding mechanism, this study investigated the changes of elements and the bond energy in the bonded region before and after using material characterization methods such as Raman spectra and XPS and analyzed the bonding mechanism. Fig. 7(b) shows the Raman spectra of Surfaces B and D. The change of the TiO_2 thin film before and after bonding can be observed by the Raman vibration frequency of the Ti–O–Ti bond, which is located around 325 cm^{-1} . The peak at this frequency is visible on Surface B but disappears on Surface D. This indicates that the TiO_2 layer disappears after laser bonding. In published papers, the Raman spectrum of CuO typically exhibits an intense peak at around 800 cm^{-1} , labeled as the F2g mode [31], [32]. Thus, in the Raman spectrum shown in Fig. 7(b), a strong peak around 800 cm^{-1} is observed, which remains constant before and after bonding. It was further found that the relative content of CuO also did not change significantly, which indicates that the bonding process maintains good gas tightness. Fig. 8(a) shows the XPS spectra of Ti corresponding to the Ti 2p orbitals of Surfaces B and D. In both Surfaces B and D, Ti–Ti, and Ti–O binding energies were evident. Nonetheless, an extra absorption peak arises at 454.3 eV on Surface D, corresponding to the Ti–Si binding energy, providing compelling evidence of Ti–Si bond formation during laser bonding. Furthermore, due to the higher electronegativity of Si than Ti (Si:1.8 and Ti:1.5 [33]), the XPS energy spectrum shows a transfer of Ti 2p to higher binding energy and Si 2p to lower binding energy [Fig. 8(b)], which is also attributable to the formation of Ti–Si bonds [34]. Fig. 8(c) and (d) shows the binding energy corresponding to the O 1s orbitals on the surface before and after bonding. The results show that the relative O content of Surfaces A and B reduced after laser bonding. This may be

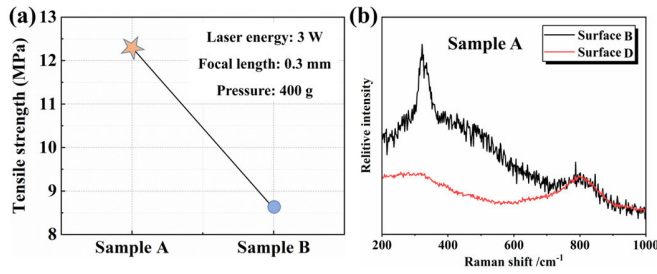


Fig. 7. (a) Comparison of the tensile strength corresponding to two surface roughness samples. (b) Raman spectra of Sample A.

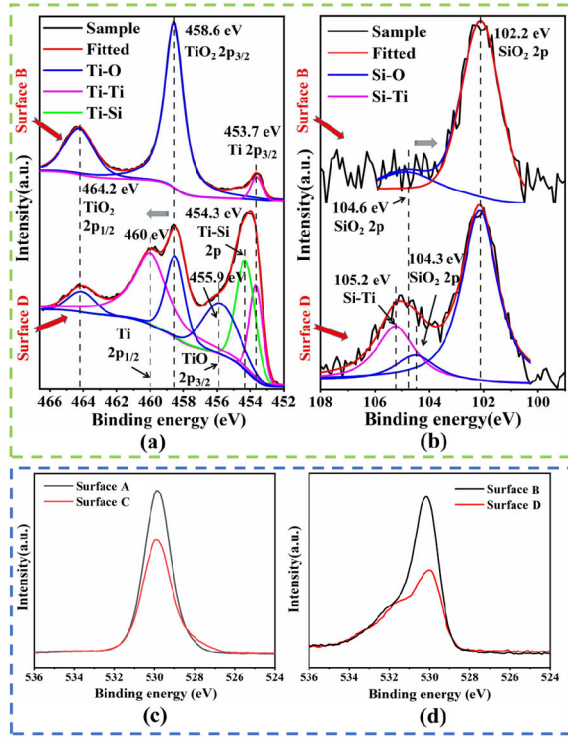


Fig. 8. XPS spectra of (a) titanium corresponding to the Ti 2p orbital and (b) Si corresponding to the Si 2p orbital of Surfaces B and D. XPS spectra of oxygen corresponding to the O 1s orbital of (c) Surfaces A and C, and (d) Surfaces B and D.

due to the interfacial temperature gradient generated by the laser energy that led to the breakage of the Si–O bond and the breakage of the Ti–O bond [9]. The rapid diffusion of Ti and Si atoms to the bonding interface led to the aggregation of Ti–Si covalent compounds (TiSi_2) with irregular interfacial morphology at high laser temperatures, which decreased the relative O content.

Bonding is a diffusion-dependent process. So, all the parameters controlling the diffusion control also the bonding's quality [18]. Based on the above characterization and analysis of the laser bonding mechanism, the laser surface modification of Ti/Cu/Ti/TiO₂ thin films in the dynamic process can be seen from Fig. 9(a). This article used borosilicate glass (BF33, dielectric constant and dielectric loss at 30 GHz are 4.8 and 0.007, respectively) manufactured by Schott with a thickness of 650 μm was used. The laser beam penetrated the glass and was directly absorbed by the TiO₂ thin film. Ti diffuses from the TiO₂ film toward the glass by the laser energy. Similarly, Si diffuses from the glass to the thin films under the

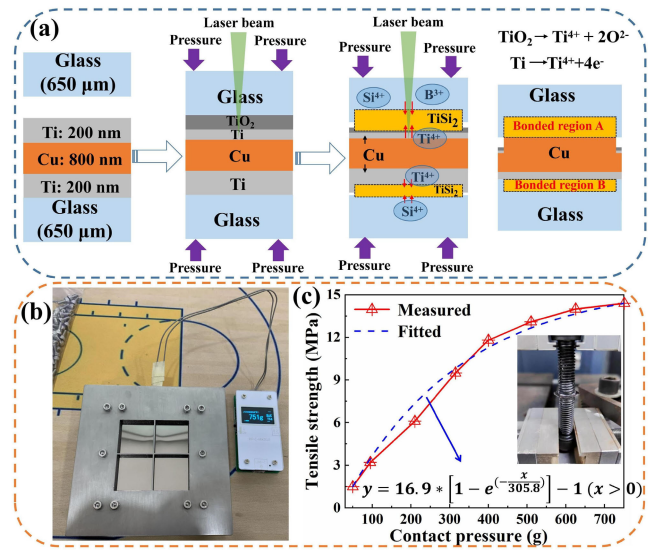


Fig. 9. Laser bonding mechanism and contact pressure analysis. (a) Schematic of the mechanism of laser surface modification with Ti/Cu/Ti/TiO₂ thin films. Contact pressure effect analysis on bonding strength (tensile strength), (b) actual laser bonding device, and (c) test results.

laser energy, forming a stable covalent structure (TiSi_2) in the bonded region when it meets Ti. Also, the heat was generated due to laser focusing on the upper TiO₂/Ti thin films and was transferred from the Cu thin film to the lower Ti thin film, causing the lower Ti thin film to be diffused by heat and form a covalent structure with the Si in the lower glass substrate. Ti moves from the oxide layer toward the glass as diffusion continues, and Si moves from the glass to the thin films' layers. As a result, the thickness of the oxide layer continued to decrease, and the oxide layer was gradually consumed. As diffusion continued, the thin oxide layer was eradicated. Also, the Ti thin film continued to provide distribution of Ti until the Ti thin film was also wholly consumed. Quantifying the XPS spectra of Ti 2p and Si 2p by integrating the peak areas reveals that the stoichiometric ratios of TiSi_2 in the bonded region A and the bonded region B are hierarchical rather than constant. In particular, there was an enrichment of TiSi_2 near the bonded region A. Ti and Si diffusion coefficients in the bonded regions A and B differed due to the laser beam path passing first through the bonded region A and then to the bonded region B, where the laser beam path creates a temperature gradient. How the laser beam path affects the composition of TiSi_2 in the bonded region A, and the bonded region B is still a question worth further investigation.

A low coefficient of expansion is necessary to ensure that the glass is not prone to breakage. According to Min et al. [19], the bonded region in the glass can be obtained to match the CTE of copper. Therefore, in this article, the CTE of borosilicate glass (BF33) was chosen to be close to that of copper while ensuring that it is as low as possible. Under these conditions, a slight increase in the coefficient of thermal expansion of the bonded region is considered suitable for a better CTE match [18].

The contact pressure effect on bonding strength is analyzed as shown in Fig. 9(b) and (c). The results show that the tensile strength increases nonlinearly with increasing contact

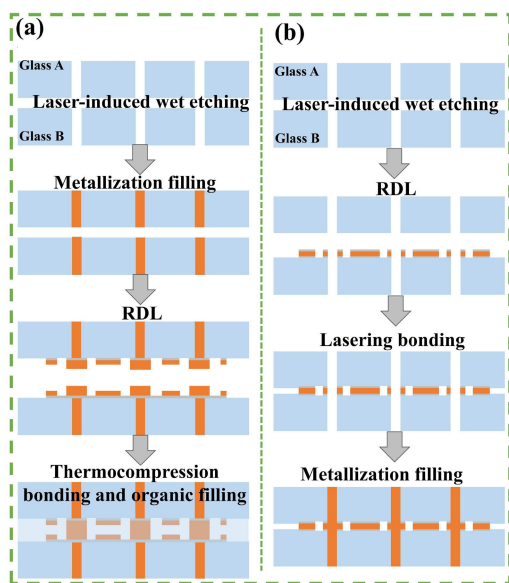


Fig. 10. Comparison of laser bonding and thermocompressive bonding processes. (a) Traditional thermocompressive bonding process in glass 3-D packaging is compared with (b) laser bonding.

pressure. At contact pressures below 400 g, the increasing trend of tensile strength with contact pressure was nearly linear. However, when it exceeded 400 g, the increasing trend of tensile strength became smaller, and after the contact pressure was 600 g, the tensile strength remained nearly constant. Primarily, the glass was crushed directly when the contact pressure was more significant than 750 g. In general, an appropriate contact pressure increases the bonding strength, as it helps to strengthen the interatomic interaction forces and increase the bonded region. However, if the applied contact pressure is too high, it may lead to deformation or fracture of the material, thus reducing the bonding strength. Therefore, the applied contact pressure must be controlled to obtain the best bonding strength during the bonding process. From fitting the test results, a functional relationship exists between tensile strength and contact pressure. The approximate tensile strength range can be predicted from the contact pressure in practical engineering. Note that minor deviations exist between the test and the fit results, which test machine errors may cause.

To better visualize the advantages of laser bonding, this article compares the laser bonding process [7] with the traditional metal thermocompressive bonding process [24], [25], [26], [27], as shown in Fig. 10. Although both methods can achieve good gas sealing and stable electrical connection. Nevertheless, laser bonding belonging to a low-temperature process, which can be performed directly at room temperature, is suitable for bonding temperature-sensitive devices. In comparison, the traditional metal thermocompressive bonding process requires heating to melt the metal [35], which may affect the performance of temperature-sensitive devices. Meanwhile, the conventional metal thermocompressive bonding process requires the metal patterns on the two bonding surfaces to be mirror symmetrical, which not only dramatically limits the design space but also has a high challenge to the alignment accuracy of the bonding [35], [36], [37], [38]. Furthermore, in terms of process cost, the laser bonding

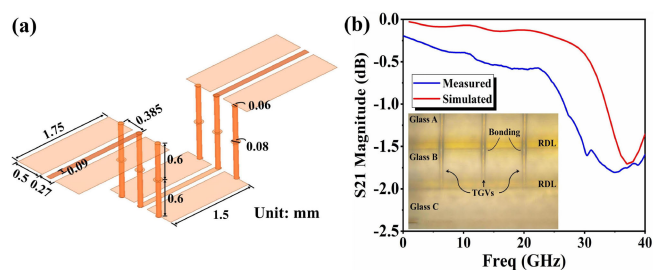


Fig. 11. (a) Full-wave simulation model of three-layer laser bonding interconnect and (b) cross-sectional microscope image of interconnect and insertion loss measured results.

TABLE I
COMPARISON WITH OTHER BONDING TECHNIQUES

Ref.	Type	Method	Tensile strength	Temp. (°C)	Bond spots	Eff.
[9]	glass/silicon	Laser bonding	9.2 MPa	500	270μm	Mode-rate
[18]	glass / Kovar-alloy	Thermo-comp.	9.84 MPa	1000	None	Poor
[23]	glass/glass	Glass frit	—	440	—	Poor
[26]	Cu/Cu	Thermo-comp.	200 MPa	140	None	Mode-rate
[29]	glass / Ti-alloy	Laser bonding	6.49 MPa	15~25	400 ~ 500μm	Mode-rate
This work	glass/glass	Laser bonding	14 MPa	15~25	None	Good

process requires the preparation of thin films on only one side of the bonding surface. In contrast, metal thermocompressive bonding involves the preparation of thin films on both sides of the bonding surfaces and requires two RDL processes [35]. Unfortunately, when thermocompressive bonds different metals, their other chemical properties and thermal expansion coefficients may create small gaps or cracks in the bonding area, reducing sealing and corrosion resistance [35], [36], [37], [38], [39]; therefore, these gaps or cracks have to be filled with organic substances to improve the sealing and corrosion resistance of the connection [27], [35]. In contrast, laser bonding can be considered free of cracks or gaps (bonding defects <0.05%). Therefore, laser bonding does not require further filling with organic materials, reducing process complexity and cost.

Table I lists the comparison of the proposed laser bonding with other bonding techniques. Compared with [9], [18], and [23], the proposed laser bonding possesses higher tensile strength and no bond spots. Moreover, the most critical thing is that no heating environment is required. However, [29] can also be applied in room temperature environments and can be used for sealing the system. However, it possesses large bond spots, limiting its application within 3-D interconnects, as large bond spots cannot be tolerated in miniaturized system packaging. Bonam et al. [26] can obtain high tensile strength with no bond spots and is widely used for 3-D packaging systems. However, as shown above, the disadvantages are the need for a heating environment and the process's complexity and high cost. Through comparative analysis, the proposed

laser bonding has certain advantages and can be one of the critical packaging technologies for glass-based microsystems.

To analyze the electrical connection performance of laser bonding, concerning [7] and [28] for the specific experimental method, the same CPW-TGVs-CPW-TGVs-CPW structure was designed, as shown in Fig. 11(a). The insertion loss of the interconnect was obtained using full-wave simulation software and microwave testing of the sample, as shown in Fig. 11(b). The results show that the multilayer interconnects achieved by laser bonding have good electrical properties and are well-suited for glass-based 3-D packaging applications.

IV. CONCLUSION

In this article, the laser modification of Ti/Cu/Ti/TiO₂ thin films was successfully used to achieve the bonding between glass and glass, and the mechanism of laser bonding was analyzed more clearly. The dynamic process of laser bonding is based on a diffusion process. The observed cross section of the sample can be divided into six regions; glass, TiSi₂, Cu-film, Ti-film, Ti/TiSi₂, and glass. The effect of surface roughness on bonding quality was investigated by comparative analysis of bonding surfaces with two samples. Meanwhile, the comparison and fitting of the tensile strength results for different contact pressures were investigated, and an exponential function relationship between contact pressure and bonding strength was found. Compared with the traditional metal thermocompressive bonding, this method can be applied to bonding temperature-sensitive devices and effectively reduce the process cost. Meanwhile, a three-layer interconnect structure was realized using this method, and the insertion loss in the range of 40 GHz was successfully measured. The results show that the insertion loss can meet the requirements of high-frequency microsystem applications. The bonding method has been used for glass-based package antenna prototypes, and excellent microwave characteristics have been obtained [7].

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